
Revealing Inaccessible Information on-the-go, by Augmenting Humans

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Abstract

People with visual impairments (PVI) experience difficulties with daily tasks that require visual queues, making PVI dependent on sighted people. Emergent assistive technology has proven to enable PVI with performing complex tasks, such as item identification by using phone applications or stand-alone assistive devices. Existing technologies currently fail to provide seamless interactions, where users can easily access information, at any location and time, without requiring the performance of multiple steps. In this research, we focus on augmenting PVI through wearable technology to provide seamless support in a mobile environment, such as the supermarket.

Author Keywords

Low vision; Wearable technology; Accessibility

ACM Classification Keywords

H.5.m [Information interfaces and presentation (e.g., HCI)]: Miscellaneous; See [<http://acm.org/about/class/1998/>]: for full list of ACM classifiers. This section is required.

Biography

Roger is a PhD student at the Augmented Human Lab, The University of Auckland, since March 2018. Roger has been working in the HCI field for the past four years at different research institutes, such as MIT Media Lab and Singapore



Figure 1: FingerReader 2.0 hardware prototype consists of a ring, incorporating a camera and a touchpad, as well as a wristband carrying a wearable computer.



Figure 2: Using a touchpad ring, we focus on evaluating a low-fidelity thumb-to-index gesture set: swipe-up/-down/-left/-right, and tap. The accuracy and latency of these gestures, as well as the applicability in running applications was studied.

University of Technology and Design. One of his main research interests is Assistive Augmentation [5], especially for people with visual impairments (PVI) Roger holds a Bachelor's degree in Electronic Engineering and has been a part of several hardware acceleration programs. He is capable of bringing prototypes to reality. His main motivation is to create a significant impact on both the research community, as well as for people with disabilities, especially PVIs.

Objectives

The main focus of this research is to substantially change how PVI access visual information on-the-go through wearable technology. To achieve this, future research needs to go beyond the typical laboratory setup and focus on real-world scenarios. Therefore, 30% of this research focuses on testing the systems we developed in the wild. On multiple occasions, a small-scale volume (10-200 units) of the latest prototype [2] were produced in order to deploy them within the blind community. By attending the *Broadening Workshop*, I aim to broaden my understanding of the research field, as well as connect with other researchers working in similar research areas. I am very open to explore and establish new meaningful collaborations.

Research Topic

People with visual impairments (PVI) face difficulties in managing daily tasks. Depending on the severity of their condition, some of their daily activities are inhibited [4], making PVI dependent on sighted people. Emergent assistive technology currently enable PVI to perform complex tasks, like Object Character Recognition [1] or Object Recognition [6], using phone applications or assistive devices [7]. Although the primary focus is to expand PVI's capabilities, most of these existing solutions are unable to provide information on scenarios that PVI most require, namely being on-the-go, while performing important tasks in

foreign environments. Furthermore, most existing devices are not intended to be wearable, as they occupy both the users' hands and require most of their senses to focus on performing a simple identification task.

The focus of this research is to augment PVIs by designing, developing and evaluating wearable technology that increases their sensory capabilities, ultimately empowering them to perform tasks on-the-go without underlying constraints from existing systems. Following the Assistive Augmentation approach[5], multiple systems capable of assisting PVI will be developed and tested. We envision these systems to also be useful in other scenarios, such as for sighted users to perform multitasking and providing assistance to the ageing community or learning support for children.

Research conducted thus far

So far, my work has contributed to three key areas:

Input: A wearable finger-worn device was developed following a human-centred-design to help PVI with performing grocery shopping by empowering the user to identify objects independently. In this work, the PVIs could point at a chosen object, while the information was provided through audio feedback (see figure 1). The prototype incorporated a micro-camera and utilized AI algorithms to identify the item the user is pointing at. The evaluation was performed in a real scenario, a supermarket, with multiple PVI (see figure 3). This work provides insights into the pointing interaction and how to present the information to the PVI. This work was accepted at IMWUT and is going to be presented at UbiComp 2018 [2]. Currently, we are exploring the limitations of camera-based assistive devices, as well as developing new systems, to overcome the existing limitations and to improve sensing capabilities.



Figure 3: Collection of pictures from FingerReader 2.0 supermarket experiment.

Interaction: To understand interaction methods in a mobile scenario using Single-hand micro-gestures (SHMGHs), a thumb-to-index interaction was implemented in a ring and evaluated with more than 20 sighted participants in a mobile context. Five different gestures were evaluated in three different mobility conditions (walking, running, standing). The research provides insights regarding the completion tasks, usability, mental task load, and enabled a greater understanding of the influence of uni-manual thumb-to-index interaction (see figure 2) in a mobile context. This work will be presented at SUI 2018 [3].

Output: Currently, I am working on developing novel AI-headphones that empowers PVLs with accessing visual information via bone-conduction headphones. A three-week evaluation process with multiple PVI was conducted, in which rich data was collected that provides valuable insights on how PVI understand items and their labelling.

In summary, these are the expected contributions of my future PhD research:

1. Understanding the needs of PVLs, as well as discovering challenges and opportunities for the design and evaluation of assistive technology.
2. Technical development of a standalone, wearable prototype to help PVI with their daily tasks.
3. Evaluation of these novel systems to understand the effect of feedback, interactions, and discover further opportunities and challenges..

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